

Realtek Wi-Fi SDK for Android JB 4.2

ver. 1.2.1

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Release History

1.0.0	2013/02/08	1. First normal release 1.1. realtek_wifi_SDK_for_android_JB_4.2_20130208.tar.gz 1.2. wpa_supplicant_8_jb_4.2_rtw_r6732.20130207.tar.gz
1.1.0	2013/03/27	1. Update wpa_supplicant_8_jb_4.2_rtw_r6985.20130327.tar.gz
1.2.0	2013/05/17	1. Update wpa_supplicant_8_jb_4.2_rtw_r7482.20130517.tar.gz
1.2.1	2013/08/21	1. Add chapter 7. FAQ and some supplementary explanations. 2. Update wpa_supplicant_8_jb_4.2_rtw_r8680.20130821.tar.gz

SDK packages

- hardware/realtek/*

Folder to store config files, private code from Realtek.

- hardware/libhardware_legacy/wifi/Android.mk
Reference codes for applying wifi_realtek.c

※ For wpa_supplicant_8_jb_4.2_rtw_r8680.20130821.tar.gz or newer version, see **wpa_supplicant_hostapd folder** of our SW release package or consult our contact window.

Introduction

This document provides a simple guide to help engineers to apply Realtek Wi-Fi solution onto their Android JB 4.2 system. For now, we have supported the following two scenarios:

- **STA/AP** Switch between STA mode and AP mode
- **(STA+P2P)/AP** Switch between STA+P2P(Wi-Fi Direct) concurrent mode and AP mode

To port Realtek Wi-Fi driver onto Android 4.2 platform, you can go through the following guide with reference codes within our driver package's realtek_wifi_SDK_for_android_JB_4.2_20130208.tar.gz.

Because Android's SDK may differ from platform to platform, our reference codes may not be applied on every platform without modifications. You should check if our reference code is suitable for you to use.

In this document, ANDROID_SDK is the path of top folder of the target Android SDK; this term is used in the following description.

1. Copy Necessary Files into SDK

After unzipping realtek_wifi_SDK_for_android_JB_4.2_20130208.tar.gz , copy following folder into the /hardware of your Android SDK.

- ANDROID_SDK /hardware/realtek

2. Platform Related Files

2.1. BoardConfig.mk

To apply Realtek Wi-Fi solution onto your Android JB system, define the following compile-time variables in BoardConfig.mk of your platform. The paths of BoardConfig.mk mostly under the following path:

ANDROID_SDK /device/<company_name>/<board_name>.

For example, BoardConfig.mk is placed under following path: ANDROID_SDK /device/ ti/panda/.

```
BOARD_WIFI_VENDOR := realtek
ifeq ($(BOARD_WIFI_VENDOR), realtek)
    WPA_SUPPLICANT_VERSION := VER_0_8_X
    BOARD_WPA_SUPPLICANT_DRIVER := NL80211
    CONFIG_DRIVER_WEXT :=y
    BOARD_WPA_SUPPLICANT_PRIVATE_LIB := lib_driver_cmd_rtl
    BOARD_HOSTAPD_DRIVER      := NL80211
    BOARD_HOSTAPD_PRIVATE_LIB := lib_driver_cmd_rtl

    BOARD_WLAN_DEVICE := rtl8192cu
    #BOARD_WLAN_DEVICE := rtl8192du
    #BOARD_WLAN_DEVICE := rtl8192ce
    #BOARD_WLAN_DEVICE := rtl8192de
    #BOARD_WLAN_DEVICE := rtl8723as
    #BOARD_WLAN_DEVICE := rtl8723au
    #BOARD_WLAN_DEVICE := rtl8189es
    #BOARD_WLAN_DEVICE := rtl8723bs
    #BOARD_WLAN_DEVICE := rtl8723bu

    WIFI_DRIVER_MODULE_NAME      := "wlan"
    WIFI_DRIVER_MODULE_PATH      := "/system/lib/modules/wlan.ko"
    WIFI_DRIVER_MODULE_ARG       := "ifname=wlan0 if2name=p2p0"

    WIFI_FIRMWARE_LOADER        := ""
    WIFI_DRIVER_FW_PATH_STA      := ""
    WIFI_DRIVER_FW_PATH_AP       := ""
    WIFI_DRIVER_FW_PATH_P2P      := ""
    WIFI_DRIVER_FW_PATH_PARAM    := ""
```

- **BOARD_WIFI_VENDOR := realtek**

To distinguish the platform Wi-Fi device from products of other companies, we define variable BOARD_WIFI_VENDOR as realtek. This is for compile-time choices to be applied for Realtek Wi-Fi solutions.

- **WPA_SUPPLICANT_VERSION := VER_0_8_X**

For Android JB, please set WPA_SUPPLICANT_VERSION as VER_0_8_X to use wpa_supplicant_8.

- **BOARD_WPA_SUPPLICANT_DRIVER := NL80211**

- **BOARD_WPA_SUPPLICANT_PRIVATE_LIB := lib_driver_cmd_rtl**

- **BOARD_HOSTAPD_DRIVER := NL80211**

- **BOARD_HOSTAPD_PRIVATE_LIB := lib_driver_cmd_rtl**

We use NL80211 as the driver interface for wpa_supplicant and hostapd to communicate with driver and provide lib_driver_cmd_rtl as the private processing library.

- **BOARD_WLAN_DEVICE**

Realtek provide a variety of Wi-Fi solutions to choose. For now, BOARD_WLAN_DEVICE is not used for any purpose but we suggest setting this variable for your Wi-Fi solution you used.

- **WIFI_DRIVER_MODULE_NAME**

- **WIFI_DRIVER_MODULE_PATH**

- **WIFI_DRIVER_MODULE_ARG**

These three variables will be used in libhardware_legacy (wifi.c/wifi_realtek.c) to do insmod and remmod. The value of WIFI_DRIVER_MODULE_NAME should match the value of MODULE_NAME specified in our driver's Makefile at compile-time. Please refer to Platform Setting Section in Detail of:

[document/Quick_Start_Guide_for_Driver_Compilation_and_Installation.pdf](#)

- **WIFI_FIRMWARE_LOADER :=""**

- **WIFI_DRIVER_FW_PATH_STA :=""**

- **WIFI_DRIVER_FW_PATH_AP :=""**

- **WIFI_DRIVER_FW_PATH_P2P :=""**

- **WIFI_DRIVER_FW_PATH_PARAM :=""**

Because our driver has FW embedded inside, and will automatically load FW at NIC initialization process, there is no need to set these 5 variables, just keep them empty.

2.2. init.xxx.rc

For Wi-Fi to operate properly, we need some daemons to be defined as service inside init.xxx.rc. In the ANDROID_SDK/device/<company_name>/<board_name>, you can see init.xxx.rc as also. For example, the path of panda board is ANDROID_SDK/device/ti/panda/init.omap4pandaboard.rc. Please add the service definitions below:

- **wpa_supplicant**

```
service rtw_suppl_con /system/bin/wpa_supplicant \
    -ip2p0 -Dnl80211 -c /data/misc/wifi/p2p_supplicant.conf -e/data/misc/wifi/entropy.bin -N \
    -iwlan0 -Dnl80211 -c/data/misc/wifi/wpa_supplicant.conf
class main
socket wpa_wlan0 dgram 660 wifi wifi
disabled
oneshot

service rtw_suppl /system/bin/wpa_supplicant -iwlan0 -Dnl80211
-c/data/misc/wifi/wpa_supplicant.conf
    socket wpa_wlan0 dgram 660 wifi wifi
class main
disabled
oneshot
```

- **dhcpcd**

```
service dhcpcd_wlan0 /system/bin/dhcpcd -aABKL
```

```
class main
```

```
disabled
```

```
oneshot
```

```
service dhcpcd_p2p /system/bin/dhcpcd -aABKL
```

```
class main
```

```
disabled
```

```
oneshot
```

```
service iprenew_wlan0 /system/bin/dhcpcd -n
```

```
class main
```

```
disabled
```

```
oneshot
```

```
service iprenew_p2p /system/bin/dhcpcd -n
```

```
class main
```

```
disabled
```

```
oneshot
```


3. System Resource Configurations

We should set the following three resource configurations of your platform to configure the network function and enable the corresponding UI interface. In general you can set the following configurations in your platform dependent config.xml file. Take panda board for example:

ANDROID_SDK/device/ti/panda/overlay/frameworks/base/core/res/res/values/config.xml

Or the global config.xml file:

ANDROID_SDK/frameworks/base/core/res/res/values/config.xml

- **NetworkAttributes**

To define the system's available network interfaces, make sure the wifi and wifi_p2p interface items are defined in the networkAttributes resource configuration in the config.xml. For example:

```
<string-array translatable="false" name="networkAttributes">
  <item>"wifi,1,1,1,-1,true"</item>
  <item>"bluetooth,7,7,0,-1,true"</item>
  <item>"ethernet,9,9,2,-1,true"</item>
  <item>"wifi_p2p,13,1,0,-1,true"</item>
</string-array>
```

- **radioAttributes**

To define the system's available network interfaces, we need to define interface items for wifi in the networkAttributes resource configuration. For example:

```
<string-array translatable="false" name="radioAttributes">
  <item>"1,1"</item>
  <item>"7,1"</item>
  <item>"9,1"</item>
</string-array>
```

- **config_tether_wifi_regexs**

The interfaces set here are used as the interfaces for Wi-Fi LAN port. We use 'wlan0' by default when our Wi-Fi is set as softap mode. So it needs to set 'wlan0' here for system to recognize 'wlan0' as Wi-Fi LAN port. For example:

```
<string-array translatable="false" name="config_tether_wifi_regexs">
  <item>"wlan0"</item>
</string-array>
```

- **config_tether_upstream_types**

The connection types set here are used as the interfaces for WAN port to connect to internet. You could declare an entry in your platform dependent config.xml file to override the global definition. For example, adding wifi and ethernet:

```
<integer-array translatable="false" name="config_tether_upstream_types">
  <item>1</item>
  <item>9</item>
</integer-array>
```

At least one item should be declared here to enable the `Tethering&portable hotspot` option of `WirelessSettings` in `Settings.apk`.

To know the definition and set other upstream connection types, please refer to `ANDROID_SDK/frameworks/base/core/java/android/net/ConnectivityManager.java`.

4. libhardware_legacy

The libhardware_legacy library includes functionality for Wi-Fi to operate. We have made modifications and extensions for our Wi-Fi solutions. To apply this, please go through the following instructions:

4.1. Apply wifi_realtek.c

Modify hardware/libhardware_legacy/wifi/Android.mk to include wifi_realtek.c instead of wifi.c into LOCAL_SRC_FILES. You also can reference the realtek_wifi_SDK_for_android_JB_4.2_20130208/hardware/libhardware_legacy/wifi/Android.mk to see how to modify Android.mk. For example:

```
ifeq ($(BOARD_WIFI_VENDOR), realtek)
LOCAL_SRC_FILES += ../realtek/wlan/libhardware_legacy/wifi/wifi_realtek.c
else
LOCAL_SRC_FILES += wifi/wifi.c
endif
```

5. wpa_supplicant_8

We provide wpa_supplicant_8_jb_4.2_rtw_r8680.20130821.tar.gz or newer version in the wpa_supplicant_hostapd/ of our SW release package. You can:

- **Compare and merge with your own wpa_supplicant_8**

Compare and merge from wpa_supplicant_8_jb_4.2_rtw by your own. For both ANDROID_SDK/external/wpa_supplicant_8/wpa_supplicant/Android.mk and ANDROID_SDK/external/wpa_supplicant_8/hostapd/Android.mk, you should notice that the two macros REALTEK_WIFI_VENDOR and ANDROID_P2P should be added into L_CFLAGS. For example:

```
ifeq ($(BOARD_WLAN_DEVICE), bcm43xx)
L_CFLAGS += -DANDROID_P2P
endif

ifeq ($(BOARD_WIFI_VENDOR), realtek)
L_CFLAGS += -DREALTEK_WIFI_VENDOR
L_CFLAGS += -DANDROID_P2P
L_CFLAGS += -DCONFIG_ANDROID_4_2_PERSISTENT_IOT
Endif
```

Here is the description of the specific macros:

MACRO	Description	
ANDROID_P2P	Android's wpa_supplicant_8 patch.	Must
REALTEK_WIFI_VENDOR	General purpose patch made by Realtek.	Must
CONFIG_ANDROID_4_2_PERSISTENT_IOT	Could decrease persistent connection time in some specific cases.	Suggest

- **Use the wpa_supplicant_8_jb_4.2_rtw instead of the original**

- A. Backup and remove the original external/wpa_supplicant_8/ folder
- B. Extract and copy the wpa_supplicant_8_jb_4.2_rtw tar file to the external/ folder of your Android SDK.
- C. Rename wpa_supplicant_8_jb_4.2_rtw as wpa_supplicant_8.

※ We have enabled the two macros ANDROID_P2P and REALTEK_WIFI_VENDOR by default.

6. Driver Configurations for Android JB

Android JB support two scenarios for Wi-Fi solution:

- **STA/AP – Switch between STA and AP mode**
- **(STA+P2P)/AP – Switch between STA+P2P concurrent and AP mode**

The configuration of driver to fit the requirement of each scenario, see the following table:

MACRO	STA /AP	(STA+P2P)/AP	Kernel ver.
CONFIG_IOCTL_CFG80211	Defined	Defined	ver. >= 2.6.35
RTW_USE_CFG80211_STA_EVENT	Defined	Defined	ver. >= 3.2.0
SUPPLICANT_RTK_VERSION_LOWER_THAN_JB42	Defined	Defined	ver. >= 2.6.35
CONFIG_CONCURRENT_MODE	Undefined	Defined	-
CONFIG_WFD	Don t Care	Must for Miracast	-
RTW_ENABLE_WIFI_CONTROL_FUNC	Must for platform device/driver mechanism		

To use RTW_USE_CFG80211_STA_EVENT on the system with kernel version between 3.0 and 3.2, please refer to the patch file:

linux-3.0.42_STATION_INFO_ASSOC_REQ_IES.diff

Refer the .diff file to modify include/net/cfg80211.h and net/wireless/nl80211.c in your kernel directory.

RTW_ENABLE_WIFI_CONTROL_FUNC is used to register platform driver callbacks. If your platform needs those callbacks, please define this macro to register platform driver callback functions. For example, these functions include:

```
static struct platform_driver wifi_device = {  
    .probe      = wifi_probe,  
    .remove     = wifi_remove,  
};
```

By default, the probe callback is used to set up Wi-Fi power and remove callback is used to close Wi-Fi power.

Please modify both the include/autoconf.h and the specific autoconf file(needed for compound driver release) for your Wi-Fi product.

Chip type	Autoconf file to modify
RTL8192CU-series	autoconf_rtl8192c_usb_linux.h
RTL8192CE-series	autoconf_rtl8192c_pci_linux.h
RTL8192DU-series	autoconf_rtl8192d_usb_linux.h
RTL8192DE-series	autoconf_rtl8192d_pci_linux.h
RTL8723AS-series	autoconf_rtl8723a_sdio_linux.h
RTL8723AU-series	autoconf_rtl8723a_usb_linux.h
RTL8723BS-series	autoconf_rtl8723b_sdio_linux.h
RTL8189ES-series	autoconf_rtl8189e_sdio_linux.h
RTL8188EU-series	autoconf_rtl8188e_usb_linux.h
RTL8723BS-series	autoconf_rtl8723b_sdio_linux.h
RTL8723BU-series	autoconf_rtl8723b_usb_linux.h

For example, if you want to configure RTL8192CU-series driver (ex: RTL8188CUS, RTL8192CU) to fit the scenario of (STA+P2P)/AP, make sure the macros: CONFIG_IOCTL_CFG80211, RTW_USE_CFG80211_STA_EVENT and CONFIG_CONCURRENT_MODE ,SUPPLICANT_RTK_VERSION_LOWER_THAN_JB42(needed for use wpa_supplicant realtek version <= jb42) in both include/autoconf.h and autoconf_rtl8192c_usb_linux.h(needed for compound driver release) are defined. As following:

```
#define CONFIG_IOCTL_CFG80211
#ifdef CONFIG_IOCTL_CFG80211
    #define RTW_USE_CFG80211_STA_EVENT
    //define CONFIG_CFG80211_FORCE_COMPATIBLE_2_6_37_UNDER
    //define CONFIG_DEBUG_CFG80211 1
    #define SUPPLICANT_RTK_VERSION_LOWER_THAN_JB42
#endif

#define CONFIG_CONCURRENT_MODE
```

Finally, we introduce two ways to compile our Wi-Fi driver, please refer to [document/Quick_Start_Guide_for_Driver_Compilation_and_Installation.pdf](#) to get more information. In above quick start guide, we introduce how to (1) compiling driver by adding or selecting target platform in Makefile (2) integrating driver source

into Linux kernel tree.

7. FAQ

7.1. Wi-Fi (STA mode)

7.1.1. Why Wi-Fi can't enable?

The whole Wi-Fi enabling procedure includes the following three main check points. Please check in sequence:

- **Is network interface(s) created?**
 - insmod driver success
 - Wi-Fi device is recognized
 - wlan0 (and p2p0) is created
- **Does wpa_supplicant run successfully?**
 - wpa_supplicant.conf (and p2p_supplicant.conf) exists and is correct
 - Service definition of wpa_supplicant exists and is correct
 - Binary file wpa_supplicant exists and is executable
- **Do connections of communication socket setup?**
 - Make sure the communication socket settings is matched below:
 - ◆ ctrl_interface in:
/data/misc/wifi/wpa_supplicant.conf
(and /data/misc/wifi/p2p_supplicant.conf)
 - ◆ Service definition of wpa_supplicant
 - ◆ Paths of communication socket in wifi.c (or wifi_realtek.c)

7.2. Portable Wi-Fi hotspot (AP mode)

7.2.1. Why Portable Wi-Fi hotspot can't enable?

The whole Portable Wi-Fi hotspot enabling procedure includes the following three main check points. Please check in sequence:

- **Is network interface created?**
 - insmod driver success
 - Wi-Fi device is recognized
 - wlan0 is created
- **Does netd and hostapd run successfully?**
 - /data/misc/wifi/hostapd.conf exists and is correct
 - Binary file netd and hostapd exist and are executable

- **Does dnsmasq run successfully?**
 - Binary file dnsmasq exist and are executable

7.3. Wi-Fi Direct (P2P mode)

7.3.1. There is no Wi-Fi Direct UI shown?

Please refer to **Add android.hardware.wifi.direct.xml** in chapter **2.3. Others** to enable Wi-Fi Direct functionality of Android 4.2.

7.3.2. Wi-Fi Direct can't scan any peer?

First, make sure you have workable Wi-Fi Direct device nearby. Make them into Wi-Fi Direct scanning state. Push **SEARCH FOR DEVICES** button also in our device and wait for a while.

If there is still no peer shown the problem is usually caused by wrong service definition of wpa_supplicant services. Please refer to **wpa_supplicant** in chapter **2.2. init.xxx.rc** to check your service definition of wpa_supplicant.